

Basic Electrical Engineering

MODULE 5 — PART 2: QUESTION BANK

Module 5 (8 hrs): Measuring Instruments, Power Transmission & Distribution, Electrical Installations

For First-Year Students — with complete answers and diagrams
Course: Basic Electrical Engineering · Course Code: EE100

Question Paper Structure

Section	Marks per Question	Scope	Questions
Section A	2 marks	Compulsory — from every module	Q1 – Q53
Section B	7 marks	Compulsory — from every module	Q1 – Q14
Section C	Numericals	Solved problems (all topics)	N1 – N10

- Every section covers **all the topics** of Module 5 — no topic is left out.
- No question is repeated between sections.
- Section C gives every numerical problem solved step by step, with its diagram.

SECTION A — 2-Mark Questions

Q1. How are electrical measuring instruments broadly classified?

Answer: Into absolute instruments and secondary instruments.

Q2. What are absolute instruments? Give an example.

Answer: Instruments that give the value of the quantity in terms of the constants of the instrument and its deflection; e.g. a tangent galvanometer. No previous calibration is needed.

Q3. What are secondary instruments?

Answer: Instruments in which the quantity is given directly by the deflection; they have a calibrated scale and are the ones most generally used.

Q4. Classify secondary instruments according to their function.

Answer: Indicating instruments (ammeter, voltmeter, wattmeter), integrating instruments (ampere-hour and watt-hour meters) and recording instruments.

Q5. What is an indicating instrument?

Answer: One that directly indicates the value of the quantity at the time of measurement, e.g. an ammeter, voltmeter or wattmeter.

Q6. What is an integrating instrument?

Answer: One that measures the total quantity of electricity (Ah) or energy (Wh) over a given time, e.g. the energy meter.

Q7. What is a recording instrument?

Answer: An indicating instrument with a pen attached to its pointer, which traces a continuous record of the variation of the quantity on a moving chart.

Q8. Name the effects used in electrical instruments.

Answer: Magnetic, electrodynamic, electromagnetic-induction, thermal, chemical and electrostatic effects.

Q9. Name the three torques required in an indicating instrument.

Answer: The deflecting (operating) torque, the controlling (restoring) torque and the damping torque.

Q10. What is the function of the deflecting torque?

Answer: It is produced by the effects of current or voltage and moves the pointer from the zero position.

Q11. What is the function of the controlling torque?

Answer: It opposes the deflecting torque and brings the pointer to rest at a position where the two are equal; it also returns the pointer to zero when the deflecting torque is removed.

Q12. What is the function of the damping torque?

Answer: It brings the pointer smoothly to its final position without oscillation, so that accurate readings can be taken quickly.

Q13. Name the methods of providing damping torque.

Answer: Air-friction damping, fluid-friction damping and eddy-current damping.

Q14. What is spring control?

Answer: A spiral hairspring provides the controlling torque; since its torsion is proportional to the twist, $T_C \propto \theta$, giving a linear scale.

Q15. What is gravity control?

Answer: A small adjustable weight provides the controlling torque; only the component $W \sin \theta$ acts, so $T_C \propto \sin \theta$ and the scale is non-uniform.

Q16. How is an ammeter connected and why does it have low resistance?

Answer: In series with the circuit; it must carry the current without increasing the circuit resistance abnormally, so its resistance is kept low.

Q17. How is a voltmeter connected and why does it have high resistance?

Answer: In parallel with the circuit; it must not draw much current or disturb the circuit, so its resistance is kept high.

Q18. On what principle does the PMMC instrument work?

Answer: On the principle that a mechanical force acts on a current-carrying conductor placed in a magnetic field.

Q19. For which supply is a PMMC instrument used and why?

Answer: For DC only, because the deflecting torque reverses with the current; on AC the average torque would be zero.

Q20. Write the deflecting-torque equation of a PMMC instrument.

Answer: $T_d = B I N A$ newton-metre, where B is the flux density, I the current, N the turns and A the coil area.

Q21. Why does a PMMC instrument have a uniform scale?

Answer: Because $T_d \propto I$ and with spring control $T_C \propto \theta$; at balance $\theta \propto I$, so the scale is linear.

Q22. State two advantages of the PMMC instrument.

Answer: Uniform scale, high accuracy and reliability, and very effective eddy-current damping.

Q23. State two disadvantages of the PMMC instrument.

Answer: It cannot be used for AC measurements and it is about 50% costlier than moving-iron instruments.

Q24. What are the two types of moving-iron instrument?

Answer: The attraction type (a single vane attracted into the coil) and the repulsion type (two vanes that repel each other).

Q25. On what principle does the moving-iron instrument work?

Answer: On the magnetic effect of current: the coil magnetises the soft-iron vanes, causing attraction or repulsion.

Q26. Why can a moving-iron instrument be used for both AC and DC?

Answer: Because reversing the current reverses the magnetisation of both vanes, so the direction of the deflecting torque is unchanged.

Q27. Why does a moving-iron instrument have a non-uniform scale?

Answer: Because the deflecting torque is proportional to the square of the current ($T_d \propto I^2$), so $\theta \propto I^2$, giving a scale crowded at the start and spread out at the end.

Q28. Name the errors in a moving-iron instrument.

Answer: Hysteresis, stray fields, temperature and friction (for both DC and AC), and frequency error (AC only).

Q29. Why is a wattmeter needed for AC power measurement?

Answer: Because AC power depends not only on voltage and current but also on the phase shift between them, $P = VI \cos \phi$.

Q30. Name the two principal types of wattmeter.

Answer: The dynamometer wattmeter (for both DC and AC) and the induction wattmeter (for AC only).

Q31. What is an energy meter?

Answer: An integrating instrument that measures electrical energy in kilowatt-hours (kWh); one unit = 1 kWh.

Q32. Name the three main parts of an electric power system.

Answer: Generation, transmission and distribution.

Q33. What is the purpose of stepping up the voltage for transmission?

Answer: To reduce the I^2R loss: for the same power, a higher voltage means a lower current, and the loss is I^2R .

Q34. What is primary transmission?

Answer: The transfer of a large quantity of power from the generating station to a substation at high voltage (about 100–700 kV) over overhead lines.

Q35. What is the distribution system made of?

Answer: Feeders, distributors and service mains.

Q36. What is a feeder?

Answer: A conductor connecting the substation to the area of distribution; no tapings are taken from it, so the current stays constant.

Q37. What is a distributor and a service main?

Answer: A distributor is a conductor from which tapings are taken for consumers; a service main is a small cable connecting the distributor to the consumer's terminals.

Q38. What is the usual low-voltage supply?

Answer: 400 V, three-phase for industrial/commercial loads and 230 V, single-phase for domestic lighting and appliances.

Q39. What is switchgear?

Answer: The apparatus used for switching, controlling and protecting electrical circuits and equipment.

Q40. What is a switch fuse unit (SFU)?

Answer: A combination of a switch (for manual isolation) and a fuse (which melts on excess current) in one unit.

Q41. On what principle does a fuse work?

Answer: On the heating effect of current: the thin fuse element melts and opens the circuit when the current exceeds its rating.

Q42. What is an MCB and what does it protect against?

Answer: A miniature circuit breaker, an automatic electromechanical switch that protects against overload and short circuit.

Q43. Give the current range of an MCB and an MCCB.

Answer: MCB: 0.5 A to 63 A; MCCB: 63 A to 3000 A.

Q44. What is an MCCB and why does it have arc extinguishers?

Answer: A moulded-case circuit breaker for large currents; interrupting a high current forms an arc, which the arc extinguishers suppress.

Q45. What is an ELCB?

Answer: An earth-leakage circuit breaker, a safety device that detects stray voltages or leakage current and interrupts the circuit to prevent electric shock.

Q46. Name any four types of wire used in electrical work.

Answer: V.I.R. wire, PVC wire, CTS/TRS wire and lead (metal) sheathed wire.

Q47. What is the difference between a wire and a cable?

Answer: A wire is a single conductor; a cable is a group of wires insulated from one another and enclosed in a protective covering.

Q48. Define earthing.

Answer: The connection of electrical machinery to the general mass of the earth through a low-resistance conductor, so that the equipment attains zero (earth) potential.

Q49. Why is earthing necessary?

Answer: To bring the equipment body to zero potential and to avoid electric shock to the operator under fault conditions.

Q50. Name the types of earthing.

Answer: Pipe earthing, plate earthing, strip (wire) earthing and rod earthing.

Q51. Classify batteries.

Answer: Primary cells (non-rechargeable, e.g. dry cell) and secondary cells (rechargeable, e.g. lead-acid, Ni-Cd, lithium-ion).

Q52. What is the capacity of a battery?

Answer: The quantity of charge it can deliver, measured in ampere-hours (Ah) = current × time of discharge.

Q53. What is the cycle life of a battery?

Answer: The number of charge-discharge cycles a secondary cell can give before its capacity falls appreciably.

SECTION B — 7-Mark Questions

Q1. Classify electrical measuring instruments and explain each class with examples.

Answer:

Instruments are divided into absolute and secondary. **Absolute instruments** give the value in terms of the constants of the instrument and its deflection (e.g. tangent galvanometer) and need no calibration; they are used only in standard laboratories. **Secondary instruments** read the value directly from a calibrated scale and are the ones in everyday use. Secondary instruments are further classified by function as indicating (ammeter, voltmeter, wattmeter), integrating (ampere-hour and watt-hour meters) and recording instruments.

Q2. Explain the essentials of an indicating instrument (the three torques).

Answer:

An indicating instrument needs three torques. The **deflecting torque** is produced by the magnetic, thermal or other effect of current and moves the pointer. The **controlling torque**, from a spring or gravity, opposes it and brings the pointer to rest where the two are equal. The **damping torque**, from air, fluid or eddy currents, brings the pointer to its final position without oscillation.

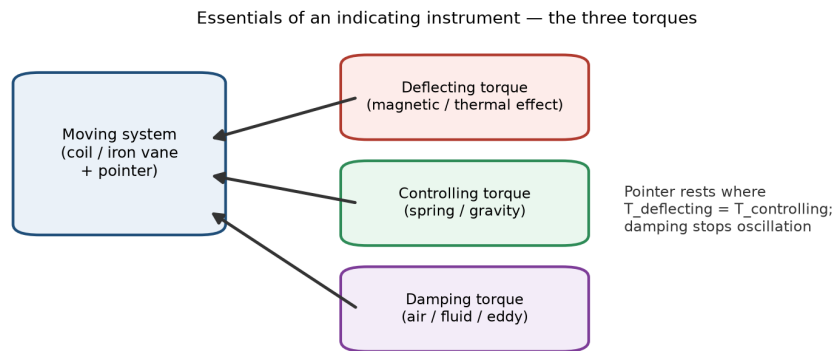


Figure 1: The three torques of an indicating instrument.

Q3. Explain spring control and gravity control with their advantages and disadvantages.

Answer:

Spring control: a phosphor-bronze hairspring twists as the pointer deflects, so $T_C \propto \theta$ and the scale is linear. Advantages: no levelling needed, springs also carry current, little extra weight. Disadvantages: temperature affects the spring, adjustment is difficult, and springs may be damaged. **Gravity control:** an adjustable weight gives $T_C = W l \sin \theta$, so the scale is non-uniform. Advantages: cheaper, unaffected by temperature, no fatigue, easily adjusted. Disadvantages: the instrument must be vertical, the weight adds to the moving system, and the scale is non-uniform.

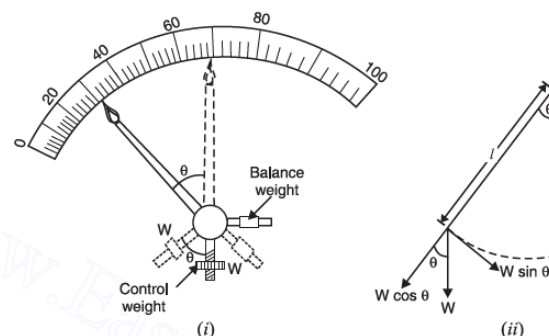


Figure 2: Gravity control.

Q4. Explain the construction and working of a PMMC instrument, and derive its deflecting torque.

Answer:

A coil is wound on an aluminium former and placed in the radial field of a permanent magnet; the pointer is attached to the moving system. When the operating current flows, a force acts on each coil side: $F = BIlN$, so the deflecting torque is $T_d = BINA$. With spring control $T_C \propto \theta$ and at balance $T_d = T_C$, giving $\theta \propto I$ and a uniform scale. Reversing the current reverses the torque, so the instrument works on DC only.

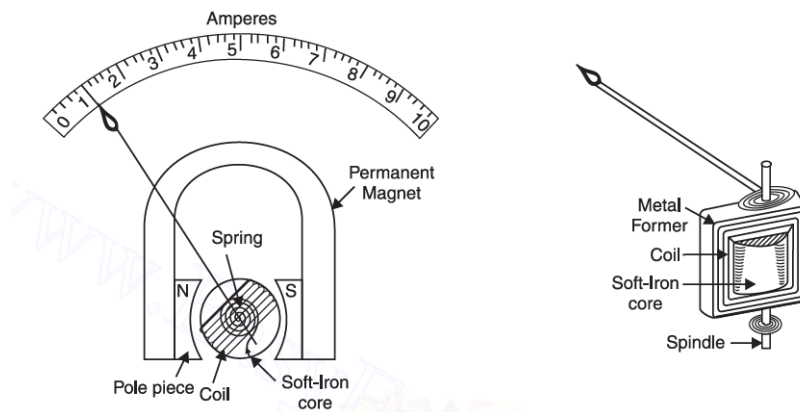


Figure 3: Construction of a PMMC instrument.

Q5. Explain the construction and working of moving-iron instruments (both types).

Answer:

In the **attraction type** a single soft-iron vane on the spindle is attracted into a fixed coil when current flows. In the **repulsion type** two vanes (one fixed, one movable) are magnetised with similar polarity and repel each other. In both, $T_d \propto I^2$, so the scale is non-uniform. Reversing the current reverses the magnetisation of the vanes, so the torque direction is unchanged — hence these instruments work on both AC and DC.

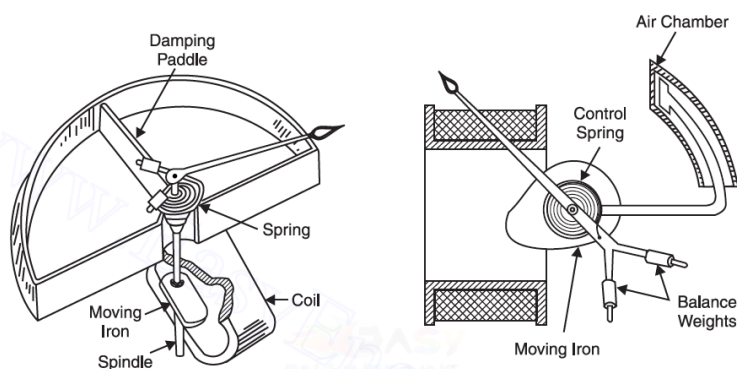


Figure 4: Attraction type.

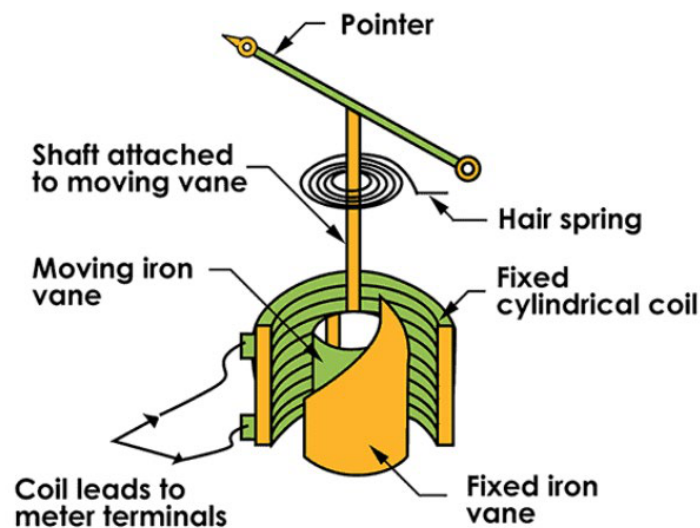


Figure 5: Repulsion type.

Q6. Explain the errors and characteristics of moving-iron instruments.**Answer:**

Errors with both DC and AC are hysteresis (readings higher when current rises), stray fields, temperature and friction; the error with AC only is due to change of frequency (corrected by a capacitor $C = L_m/R_s^2$). Advantages: cheap, robust, simple, work on AC and DC, high operating torque. Disadvantages: non-linear scale, less sensitive than PMMC, and frequency error on AC.

Q7. Explain the dynamometer wattmeter with its advantages and disadvantages.**Answer:**

It works on the dynamometer principle — a mechanical force between two current-carrying coils. The fixed current coil is in series with the load and the movable potential coil is in parallel through a high resistance. The deflection is proportional to $V I \cos \phi$, so it reads real power. It can be used for both AC and DC, has a uniform scale and gives high accuracy; but at low power factor the potential-coil inductance causes errors, stray fields affect the reading, and the coils may burn out if the ratings are exceeded.

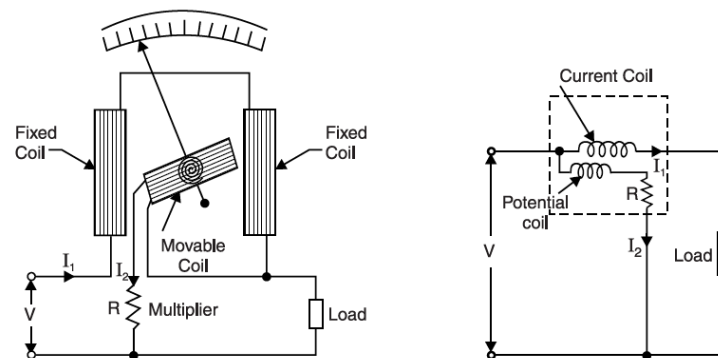


Figure 6: Dynamometer wattmeter.

Q8. Explain the structure of an electric power system.**Answer:**

An electric power system has three parts. **Generation** converts fuel into electrical energy, usually at 11–25 kV. **Transmission** moves bulk power over long distances; the voltage is stepped up (primary transmission, 100–700 kV) to reduce I^2R loss and later stepped down (secondary transmission, 33–66 kV, then 11 kV). **Distribution** delivers power to local consumers through feeders, distributors and service mains.

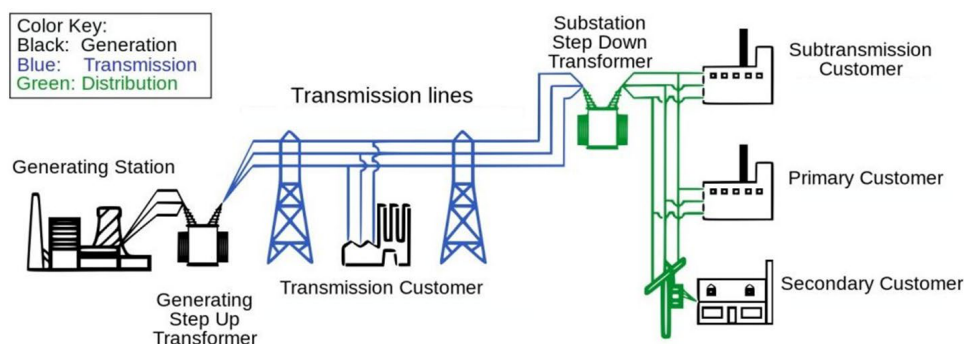


Figure 7: Structure of an electric power system.

Q9. Explain generation, primary transmission and secondary transmission with typical voltages.**Answer:**

Generation takes place at 11–25 kV in thermal, hydro or nuclear plants. In primary transmission the voltage is stepped up to 100–700 kV (higher for longer distances) to reduce the I^2R loss, and the power is carried to a receiving station over overhead lines. In secondary transmission the voltage is stepped down to 33–66 kV and sent to substations near the load centres, then to about 11 kV for distribution.

Q10. Explain the distribution system and the low-voltage distribution (400 V / 230 V).**Answer:**

The distribution system lies between the substation and the consumers' meters and consists of feeders, distributors and service mains. At the distribution transformer the voltage is brought down to the low-voltage level: 400 V three-phase (3-phase, 4-wire) for motors and commercial loads, and 230 V single-phase (phase to neutral) for domestic lighting and appliances.

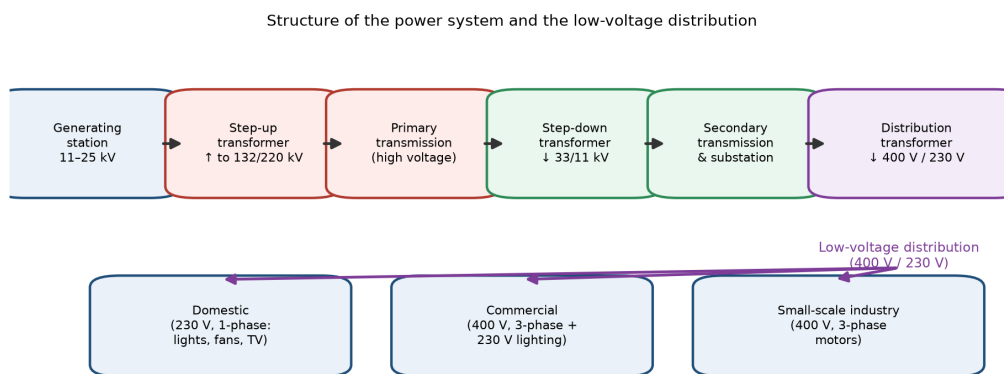


Figure 8: Low-voltage distribution (400 V / 230 V).

Q11. Explain the components of LT switchgear: SFU, MCB, MCCB and ELCB.**Answer:**

The **SFU** combines a manually operated switch (for isolation) with a fuse that melts on excess current. The **MCB** automatically makes and breaks the circuit and trips on overload or short circuit; it is rated 0.5–63 A. The **MCCB** is similar but for 63–3000 A, uses a thermal (bimetallic) mechanism and has arc extinguishers. The **ELCB** detects leakage current or stray voltage and interrupts the circuit to prevent electric shock.



Switch fuse unit

Figure 9: Switch fuse unit.



Figure 10: Miniature circuit breaker.

Q12. Explain the types of wires and cables used in electrical installations.

Answer:

A wire is a single conductor; a cable is a group of insulated wires with a protective covering. Common wires are V.I.R. (rubber-insulated, for dry places), PVC (tough, moisture-proof, most common), CTS/TRS (rubber-sheathed, for damp climates) and lead/metal sheathed (for mechanical and moisture protection, earthed at junctions). Cables may be single-core or multi-core and armoured (steel covering, for underground use) or unarmoured.

Q13. Explain earthing, its necessity and its types.**Answer:**

Earthing is the connection of electrical equipment to the general mass of the earth through a low-resistance conductor, so that the equipment body reaches zero (earth) potential. It is necessary to avoid electric shock to the operator when insulation fails and a live part touches the casing. The types are pipe earthing (a GI pipe buried vertically), plate earthing (a copper or GI plate buried vertically), strip/wire earthing (strip electrodes in trenches, for rocky terrain) and rod earthing (a rod driven vertically, where soil resistivity is low).

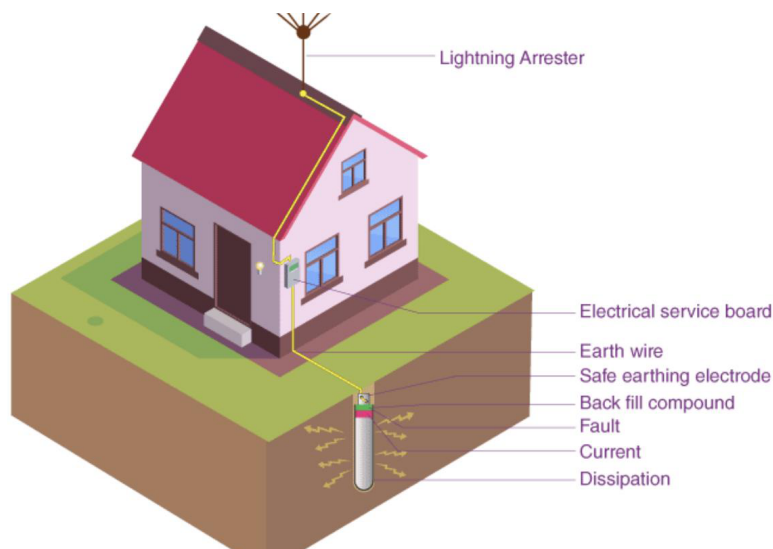


Figure 11: Earthing of equipment.

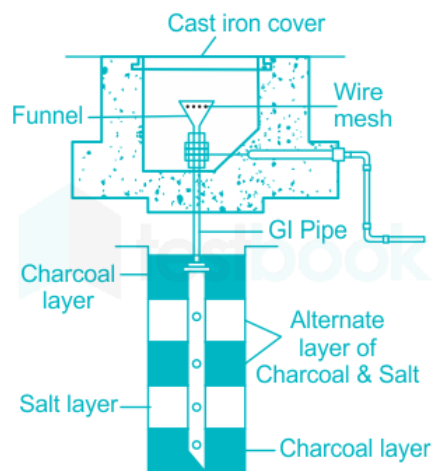


Figure 12: Pipe earthing.

Q14. Explain the types of batteries and their important characteristics.**Answer:**

Batteries are of two kinds. **Primary cells** are non-rechargeable (dry cell, alkaline cell); **secondary cells** are rechargeable (lead-acid, nickel-cadmium, lithium-ion). Important characteristics are capacity (in Ah), voltage (about 2 V per lead-acid cell), energy density, internal resistance, cycle life, self-discharge, efficiency and specific gravity.

SECTION C — Numerical Problems (Solved)

Every numerical below is solved step by step.

N1. Shunt Resistance for an Ammeter

Problem: A PMMC meter of resistance $100\ \Omega$ gives full-scale deflection at $10\ \text{mA}$. Find the shunt resistance needed to use it as a $1\ \text{A}$ ammeter.

Solution:

$$R_{sh} = R_m I_m / (I - I_m).$$

$$R_{sh} = 100 \times 0.01 / (1 - 0.01) = 1 / 0.99 = 1.01\ \Omega.$$

Answer: $R_{sh} = 1.01\ \Omega$.

N2. Multiplier Resistance for a Voltmeter

Problem: The same meter ($100\ \Omega$, $10\ \text{mA}$ full scale) is to be used as a $100\ \text{V}$ voltmeter. Find the series multiplier resistance.

Solution:

$$R_s = V / I_m - R_m.$$

$$R_s = 100 / 0.01 - 100 = 10000 - 100 = 9900\ \Omega.$$

Answer: $R_s = 9900\ \Omega$.

N3. Deflecting Torque of a PMMC Instrument

Problem: A PMMC instrument has a coil of 100 turns and area $2 \times 10^{-3}\ \text{m}^2$ in a field of flux density $0.1\ \text{Wb/m}^2$. If a current of $10\ \text{mA}$ flows, find the deflecting torque.

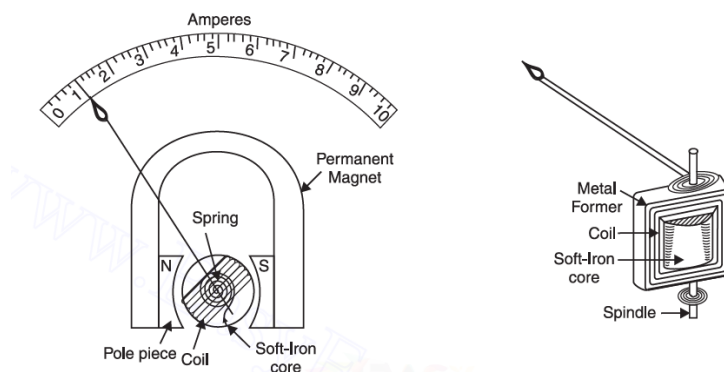


Figure 13: Figure for N3.

Solution:

$$T_d = B I N A.$$

$$T_d = 0.1 \times 0.01 \times 100 \times 2 \times 10^{-3} = 2 \times 10^{-4}\ \text{N}\cdot\text{m}.$$

Answer: $T_d = 2 \times 10^{-4}\ \text{N}\cdot\text{m}$.

N4. Energy Consumed by a Load

Problem: A load of $1.5\ \text{kW}$ runs for $8\ \text{hours}$. Find the energy consumed in kWh and its cost at ₹5 per unit.

Solution:

$$\text{Energy} = \text{Power} \times \text{time} = 1.5\ \text{kW} \times 8\ \text{h} = 12\ \text{kWh}.$$

$$\text{Cost} = 12\ \text{units} \times ₹5 = ₹60.$$

Answer: Energy = $12\ \text{kWh}$; cost = ₹60.

N5. Energy Consumed by Several Appliances

Problem: Five lamps of 40 W each run for 6 hours and a 1 kW heater runs for 2 hours. Find the total energy consumed.

Solution:

Lamp energy = $5 \times 40 \times 6 = 1200 \text{ Wh} = 1.2 \text{ kWh}$.

Heater energy = $1000 \times 2 = 2000 \text{ Wh} = 2 \text{ kWh}$.

Total = $1.2 + 2 = 3.2 \text{ kWh}$.

Answer: Total energy = 3.2 kWh (units).

N6. Reading of an Energy Meter

Problem: An energy meter makes 1200 revolutions per kWh. How much energy is consumed if the disc makes 600 revolutions?

Solution:

Energy = revolutions / (revolutions per kWh).

Energy = $600/1200 = 0.5 \text{ kWh}$.

Answer: Energy = 0.5 kWh.

N7. Wattmeter Reading

Problem: A wattmeter is connected to a load drawing 10 A at 230 V with a power factor of 0.8 lagging. Find the power indicated.

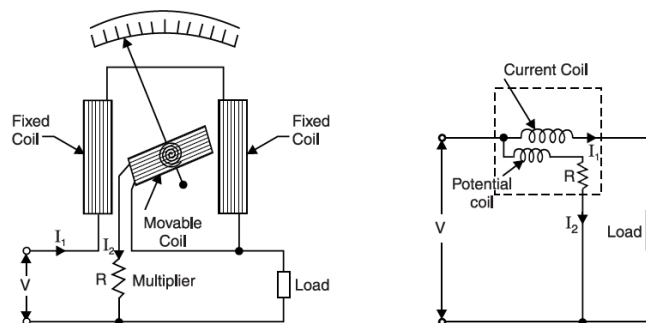


Figure 14: Figure for N7.

Solution:

$P = V I \cos \phi$.

$P = 230 \times 10 \times 0.8 = 1840 \text{ W}$.

Answer: $P = 1840 \text{ W}$.

N8. Battery Capacity

Problem: A battery delivers a steady current of 5 A for 4 hours. Find its capacity and the energy it supplies at 12 V.

Solution:

Capacity = $I \times t = 5 \times 4 = 20 \text{ Ah}$.

Energy = $V \times \text{capacity} = 12 \times 20 = 240 \text{ Wh} = 0.24 \text{ kWh}$.

Answer: Capacity = 20 Ah; energy = 240 Wh.

N9. Battery Energy from Ah Rating

Problem: A 12 V, 100 Ah battery is fully charged. Find the energy stored in kWh.

Solution:

Energy = Voltage $\times$ Capacity.

Energy = $12 \times 100 = 1200 \text{ Wh} = 1.2 \text{ kWh}$.

Answer: Energy = 1.2 kWh.

N10. Why Transmission Voltage is Stepped Up

Problem: A power of 10 MW is to be transmitted through a line of resistance 10Ω . Find the I^2R loss at (i) 11 kV and (ii) 132 kV, and comment.

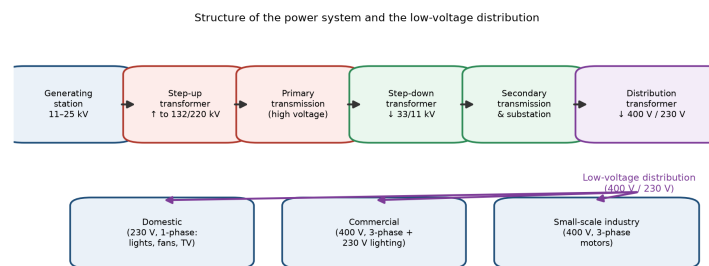


Figure 15: Figure for N10.

Solution:

At 11 kV: $I = P/V = 10 \times 10^6 / 11000 = 909 \text{ A}$; loss = $I^2R = 909^2 \times 10 = 8.26 \text{ MW}$ (very large).

At 132 kV: $I = 10 \times 10^6 / 132000 = 75.76 \text{ A}$; loss = $75.76^2 \times 10 = 57.4 \text{ kW}$ (very small).

Thus, stepping up the voltage greatly reduces the I^2R loss, which is why power is transmitted at high voltage.

Answer: Loss = 8.26 MW at 11 kV and 57.4 kW at 132 kV.